This combination of gunpowder technology with field fortifications made Mughal tactics truly devastating. Moreover, gunpowder technology was not adopted blindly or haphazardly. The Mughals, led by Akbar himself, were obsessed with guns and their uses and tinkered endlessly with them. Akbar devised new ways to use this musket and cannon technology in response to the demands of the empire and the challenges he faced. So, for example, Akbar devised rockets,* light and easy to transport. Rockets were found to be particularly effective against enemy animals—horses and elephants, for their whistling shriek caused pandemonium among them. The Mughals then devised a special rocket with an in-built whistle to make better use of this effect and, as pointed out by Andrew de la Garza, 'rocketry is one instance where the West adopted Indian military technology'. The Mughals adapted swivel camel guns and elephant guns, high-performing muskets, lethal field artillery of different sizes all in addition to the famed horse archer. 'The resulting method of warfare would appear alien to Western observers', used to linear formations and close order drill:

> …but it was in reality a logical response to an extremely hazardous battlefield environment dominated by fire. It actually demanded a higher degree of discipline and initiative from the individual soldier and small unit commander and in many ways anticipated much later developments in Europe after the introduction of rifle muskets, breechloaders and even more lethal field artillery.

Meanwhile Akbar's sudden show of aggression in Surat in 1580 seems to have been at least partly due to more personal calculations. Though the Portuguese had indeed proved to be a constant menace, it was the coastal trading kingdoms of the south who had been most affected. The Mughals, with their focus directed towards the huge agricultural heartland, were only minimally inconvenienced. But from the time that Gulbadan and her party of Mughal women had planned to travel to Mecca, Akbar had been supremely solicitous of their safety. As we have seen, he had ensured friendly relations with the Portuguese at Surat in 1573 so that a cartaz was readily obtained. Despite this, Gulbadan and her party had been delayed for a year in Surat while the authorities prevaricated and Gulbadan had offered the Portuguese the town of Baksar to expedite matters. Then in 1577 there were reports of the Portuguese capturing Gujarati ships returning from